



RETRIEVAL & SEARCH

RAG Systems & Vector Databases

Enterprise retrieval-augmented generation with hybrid semantic search, chunking, and re-ranking.

EXECUTIVE SUMMARY

Effective RAG systems require more than simple top-k vector cosine similarity. This architecture guide details hybrid search (dense embeddings + BM25 lexical), semantic document chunking, and cross-encoder re-ranking.

1 Hybrid Search & Retrieval Architecture

Combining dense vector embeddings with sparse keyword search (BM25) ensures accurate retrieval for both high-level semantic meaning and exact technical terms (serial numbers, IDs, and code symbols).



2 Production Implementation & Schema

```
-- Hybrid Search Schema with pgvector in Neon Postgres
CREATE TABLE document_chunks (
  id SERIAL PRIMARY KEY,
  content TEXT NOT NULL,
  embedding vector(1536),
  tsv_content TSVECTOR GENERATED ALWAYS AS (to_tsvector('english', content)) STORED
);
CREATE INDEX ON document_chunks USING hnsw (embedding vector_cosine_ops);
```



3 Reciprocal Rank Fusion (RRF) & Re-Ranking

RRF merges ranked results from vector and keyword search queries without needing score normalization, consistently delivering higher answer relevance than pure vector lookups.

rag-benchmark · Retrieval Accuracy

CONNECTED

```
[BENCHMARK] Evaluating 500 enterprise knowledge queries...
[PURE VECTOR] Recall@5: 72.4% (struggled with exact acronyms)
[PURE BM25] Recall@5: 68.1% (struggled with conceptual queries)
✓ [HYBRID + RRF] Recall@5: 91.8% · Significant accuracy improvement
[SYNTHESIS] 0 hallucinated facts across validated ground-truth tests
```

4 Key Takeaways & Best Practices

- ✓ Semantic Chunking

Break documents by natural section headings rather than arbitrary character counts to preserve sentence context.
- ✓ Hybrid Indexing

Always run pgvector alongside PostgreSQL full-text search for optimal precision on both concepts and keywords.
- ✓ Re-Ranking Step

Pass the top 20 candidate chunks through a cross-encoder re-ranker before submitting the final 5 to the LLM.
- ✓ Grounded Citations

Instruct the generator model to include chunk IDs with each claim to enable instant human auditing.

IMPLEMENTATION SUPPORT

Need engineering assistance integrating these architectures into your business? Contact our team directly at futureai4all@gmail.com.